THRIFTKART / SEMESTER III

Software Requirements Specification

IEEE 830-style requirements baseline

System	ThriftKart Web Marketplace
Version	1.0 / Assessment baseline
Requirement set	12 Functional + 8 Non-functional
Priority	MoSCoW
Traceability	FR → Test Case
Source basis	Case Study no. 72 + product design decisions

1. Introduction

1.1 Purpose

This SRS defines the functional behavior, measurable quality requirements, interfaces, data requirements, business rules and acceptance criteria for ThriftKart. It is written as the implementation baseline for the campus textbook marketplace and the evidence package for the Software Engineering & Project Management assessment.

1.2 Product Scope

ThriftKart supports student account access, second-hand textbook listings, moderation, marketplace search, buyer-seller communication, purchase requests, transaction/payment status and campus handover. The product intentionally avoids delivery logistics and a full payment gateway in the MVP.

1.3 Definitions

Term	Meaning
Listing	A textbook offer created by a seller.
Moderation	Admin review that determines whether a listing may be publicly visible.
Reservation	The active purchase hold placed against an available listing.
Transaction	The record connecting buyer, seller and listing through the sale/handover lifecycle.
Handover	The physical campus exchange between buyer and seller.

2. Overall Description

2.1 Product Perspective

The MVP is a responsive web application backed by a relational data model. The web client communicates with an application/API layer; the API owns validation and state transitions; persistent data is stored in a relational database. Payment is represented as a status/audit module so the marketplace does not become coupled to a specific external gateway.

2.2 User Classes

User class	Capabilities
Student buyer	Search, filter, view, chat, request purchase, confirm handover, view transaction status.
Student seller	Create/edit listing, submit for moderation, respond to buyer, confirm handover, view transaction status.
Admin moderator	Review listings, approve/reject, review reports, inspect audit evidence.
System operator / deployment admin	Deploy application, configure environment, monitor health; no business workflow ownership.

2.3 Operating Environment

Layer	Baseline
Client	Modern desktop/mobile browser; responsive 320px–1440px design target.
Frontend	HTML + CSS + vanilla JavaScript prototype; implementation may later use a JS framework.
Backend	REST-style API boundary; any implementation language/framework may be used if it preserves requirements.
Database	Relational database with enforced foreign keys and indexed marketplace fields.
Hosting	Any web hosting arrangement appropriate to the student MVP; the case does not prescribe a provider.

3. Functional Requirements

ID	Feature	Requirement	MoSCoW
FR-01	Account access	The system shall allow a student to register and sign in using a university email and password.	Must
FR-02	Profile management	The system shall allow authenticated students to view and update their name, study year and contact preference.	Should
FR-03	Create listing	The system shall allow a seller to create a draft textbook listing with title, author, course, edition, price, condition, description and campus handover point.	Must
FR-04	Listing validation	The system shall validate required fields, price limits and the 1-5 condition rating before accepting a listing for moderation.	Must
FR-05	Moderation workflow	The system shall place submitted listings into Pending Moderation and allow an admin moderator to approve or reject them with a reason.	Must
FR-06	Marketplace search	The system shall allow users to search published listings and filter by category, condition and price range.	Must
FR-07	Listing detail	The system shall show a listing detail view containing book information, seller identity summary, price, condition and handover location.	Must
FR-08	Buyer-seller chat	The system shall allow a buyer to start and continue a conversation with the seller for a specific listing.	Should
FR-09	Purchase request	The system shall allow a buyer to reserve/request a purchase for an available listing and prevent conflicting active reservations.	Must
FR-10	Payment status	The system shall record a transaction payment method and status (for example Cash on Handover, External UPI, Pending, Confirmed) without requiring a full payment gateway in the MVP.	Must
FR-11	Campus handover	The system shall allow buyer and seller to record the agreed campus handover point/time and mark the transaction as completed after handover.	Must
FR-12	Trust and safety	The system shall allow reporting of a listing/user and provide an admin queue to review reports and moderate offending content.	Should

4. Non-Functional Requirements

ID	Quality	Measurable requirement	Category
NFR-01	Performance	For search, filter and listing-detail requests, at least 95% of responses shall complete within 2 seconds under the pilot load profile.	Performance
NFR-02	Availability	The pilot system shall achieve at least 99.0% monthly availability outside planned maintenance.	Availability
NFR-03	Usability	At least 90% of first-time student testers shall complete the core task of finding a book or submitting a listing without facilitator intervention.	Usability
NFR-04	Trust & safety	100% of new listings shall enter Pending Moderation before appearing in the public marketplace; 100% of moderation decisions shall store moderator, time and decision reason.	Trust & Safety
NFR-05	Security	100% of protected API routes shall reject unauthenticated requests, and no password shall be stored in plaintext.	Security
NFR-06	Data integrity	100% of completed transactions shall reference valid buyer, seller and listing records enforced by database foreign keys.	Integrity
NFR-07	Responsive UX	All core flows shall remain usable from 320px to 1440px viewport widths with no horizontal page scrolling.	Usability
NFR-08	Testability	All critical price and moderation rules shall have executable test cases, and each FR shall map to at least one named test case in the traceability matrix.	Maintainability / QA

5. External Interfaces

Interface	Requirement
University authentication	MVP may use application-managed credentials with university email validation; any future SSO integration shall preserve FR-01 semantics.
Payment methods	External UPI or cash-on-handover may be recorded as a method/status; no gateway is required for MVP.
Email / notification	Optional for MVP; the system may display in-app transaction state instead of building a notification pipeline.
Image storage	Listing images shall be stored as URLs/objects referenced by listing_images; exact provider is implementation-dependent.

6. Data Requirements

Table	Key attributes	Relationship / rule	Purpose
users	user_id, university_email, password_hash, full_name, year_of_study, role, status, created_at	PK user_id; UNIQUE university_email	Student/admin identity and access
categories	category_id, name, slug	PK category_id; UNIQUE slug	Textbook grouping
listings	listing_id, seller_id, category_id, title, author, isbn, edition, course_code, price, condition_rating, description, campus_location, status, created_at, updated_at	PK listing_id; FK seller_id -> users; FK category_id -> categories	Core textbook marketplace record
listing_images	image_id, listing_id, image_url, sort_order	PK image_id; FK listing_id -> listings	One-to-many listing media
conversations	conversation_id, listing_id, buyer_id, seller_id, status, created_at	PK conversation_id; FKs to listings/users; UNIQUE(listing_id,buyer_id,seller_id)	Buyer-seller conversation per listing
messages	message_id, conversation_id, sender_id, body, sent_at, read_at	PK message_id; FK conversation_id -> conversations; FK sender_id -> users	Chat messages
transactions	transaction_id, listing_id, buyer_id, seller_id, agreed_price, payment_method, payment_status, handover_location, handover_at, status, created_at, completed_at	PK transaction_id; FKs to listings/users	Sale lifecycle and campus handover
payment_records	payment_id, transaction_id, amount, method, provider_reference, status, paid_at	PK payment_id; FK transaction_id -> transactions	Payment confirmation/audit record; gateway optional
moderation_reviews	review_id, listing_id, moderator_id, decision, reason, reviewed_at	PK review_id; FK listing_id -> listings; FK moderator_id -> users	Approval/rejection evidence
reports	report_id, reporter_id, listing_id, reported_user_id, reason, status, created_at, resolved_at	PK report_id; FKs to users/listings	Trust and safety reporting queue

Schema note

The table set above is a proposed implementation model for the SRS. The supplied case study does not prescribe database table names, so these are design artefacts derived from the required marketplace and moderation workflows.

7. Business Rules

ID	Rule	Definition
BRULE-01	University email required	Only verified university-domain accounts can use buyer/seller flows.
BRULE-02	Moderation gate	A listing cannot be public unless its moderation decision is Approved.
BRULE-03	Price validation	Proposed product rule: price must be a positive whole-number amount up to ₹20,000.
BRULE-04	Condition validation	Condition rating must be an integer from 1 (poor) to 5 (excellent).
BRULE-05	Single active reservation	A listing may have at most one active reservation at a time.

ID	Rule	Definition
BRULE-06	Campus handover	Completed transactions require a handover point and a completion confirmation.
BRULE-07	Auditability	Every moderation action stores moderator, decision, reason and timestamp.

8. Core Use Case Specifications

ID	Use case	Actor(s)	Outcome
UC-01	Register / Sign In	Student	Valid university email + password → authenticated session.
UC-02	List a Book	Seller	Enter data → validate → save draft → submit → Pending Moderation.
UC-03	Moderate Listing	Admin	Open queue → inspect → approve/reject → record reason.
UC-04	Find a Book	Buyer	Search/filter → open listing → inspect seller + condition + price.
UC-05	Complete a Sale	Buyer + Seller	Request → optional chat → payment status → handover → complete transaction.
UC-06	Report / Resolve	Student + Admin	Create report → moderation queue → decision → resolution status.

9. Traceability Matrix

Requirement	Primary test case	Evidence focus
FR-01	TC-01	Account access
FR-02	TC-02	Profile management
FR-03	TC-03	Listing creation
FR-04	TC-04	Price + condition validation
FR-05	TC-05	Moderation decision
FR-06	TC-06	Search/filter
FR-07	TC-07	Listing detail
FR-08	TC-08	Chat
FR-09	TC-09	Single active reservation
FR-10	TC-10	Payment status
FR-11	TC-11	Campus handover
FR-12	TC-12	Report + admin queue

10. Acceptance Criteria

- No listing reaches Published without an Approved moderation decision.
- Price and condition validation rejects values outside the defined product boundaries.
- Search results can be filtered and a selected listing shows enough information to decide whether to request it.
- Only one active reservation may exist for a listing at a time.
- A completed sale records payment status and campus handover details.
- All eight NFRs have measurable targets and can be validated by the test plan.

11. Assumptions, Constraints & Exclusions

Type	Statement
Assumption	Students have campus access and can arrange face-to-face handover.
Assumption	The MVP can use simple account access instead of enterprise SSO.
Constraint	The team has 4 students × 3 productive hours/day on weekdays and no capacity in weeks 9–10.
Exclusion	Delivery logistics and complex payment gateway integration are outside the MVP.
Exclusion	Advanced recommendation/personalization is not required to satisfy the case.